LABSHEET -9 COMPUTER VISION

DHATRI KASHYAP CU240250904

```
!pip install ultralytics
```

Show hidden output

```
# Install Ultralytics
!pip install -q ultralytics
# Import libraries
import cv2
import numpy as np
import matplotlib.pyplot as plt
import os
from ultralytics import YOLO
print("All libraries imported successfully!")
```

```
Creating new Ultralytics Settings v0.0.8 file ✅
View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see https://docs.ultral
All libraries imported successfully!
```

```
# 1. LOAD PRE-TRAINED YOLOv8 MODEL
# ============================================================

print("\nLoading YOLOv8 model...")

model = YOLO("yolov8n.pt")

print("YOLOv8 model loaded successfully!")
```

```
Loading YOLOv8 model...
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolov8n.pt to 'yolov8n.pt': 100% ━━━━━━━━━━━━ 6.
YOLOv8 model loaded successfully!
```

```
# 2. UPLOAD IMAGE
from google.colab import files

uploaded = files.upload()

if len(uploaded) == 0:
    print("ERROR: No image was uploaded.")
else:
    # Get the actual uploaded file name
    file_name = list(uploaded.keys())[0]
    image_path = "/content/" + file_name

    print("\nImage uploaded successfully!")
    print("Image Path:", "/content/pexels-friededia-30649280.jpg")
```

```
Choose Files pexels-fried...649280.jpg
pexels-friededia-30649280.jpg(image/jpeg) - 5622111 bytes, last modified: 8/13/2026 - 100% done
Saving pexels-friededia-30649280.jpg to pexels-friededia-30649280 (1).jpg

Image uploaded successfully!
Image Path: /content/pexels-friededia-30649280.jpg
```

```
# 3. LOAD IMAGE
image = cv2.imread("/content/pexels-friededia-30649280.jpg")
if image is None:
    print("ERROR: Image could not be loaded.")
else:
    print("\nImage loaded successfully!")
    print("Image Size:", image.shape)
```

```
Image loaded successfully!
Image Size: (5464, 8192, 3)
```

```
# 4. IMAGE PREPROCESSING

import cv2

# Convert BGR image to RGB
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
```

```
# Resize image to 640 x 640
processed_image = cv2.resize(image_rgb, (640, 640))

print("Image preprocessing completed.")
print("Image Size:", processed_image.shape)
print("PRIYANSHUN SHARMA CU240250980")
```

```
Image preprocessing completed.
Image Size: (640, 640, 3)
PRIYANSHUN SHARMA CU240250980
```

```
%pip install -q ultralytics
```

```
# 5. PERFORM OBJECT DETECTION
# ============================================================

print("\nDetecting objects...")
print("PRIYANSHUN SHARMA CU240250980")

results = model(image_path)

print("Object detection completed.")
```

```
Detecting objects...
PRIYANSHUN SHARMA CU240250980

image 1/1 /content/pexels-friededia-30649280 (1).jpg: 448x640 (no detections), 437.0ms
Speed: 16.5ms preprocess, 437.0ms inference, 16.8ms postprocess per image at shape (1, 3, 448, 640)
Object detection completed.
```

```
# 6. GET DETECTION RESULT
result = results[0]
# Create image with bounding boxes
detected_image = result.plot()
# Convert BGR to RGB for Matplotlib
detected_image_rgb = cv2.cvtColor(
    detected_image,
    cv2.COLOR_BGR2RGB
)
```

```
# 7. PRINT DETECTION RESULTS
print("\n============================================================")
print("DETECTED OBJECTS")
print("============================================================")
print("PRIYANSHUN SHARMA CU240250980")

object_count = 0

for box in result.boxes:

    object_count += 1

    # Class ID
    class_id = int(box.cls[0])

    # Object name
    class_name = model.names[class_id]

    # Confidence score
    confidence = float(box.conf[0])

    # Bounding box coordinates
    x1, y1, x2, y2 = box.xyxy[0].tolist()

    print("\nObject", object_count)

    print("Class       :", class_name)

    print("Confidence  :", round(confidence, 2))

    print(
        "Bounding Box:",
        "(",
        round(x1, 2),
        ",",
        round(y1, 2),
        ",",
        round(x2, 2),
        ",",
```

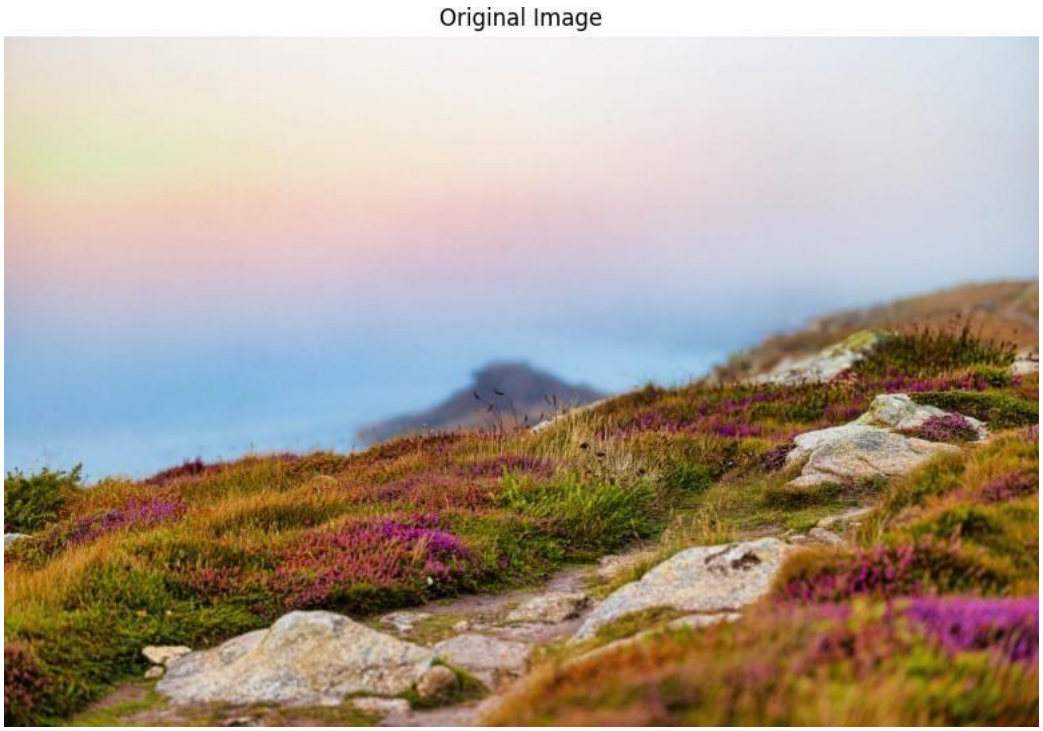
```
            round(y2, 2),
            ")"
        )
```

```
============================================================
DETECTED OBJECTS
============================================================
PRIYANSHUN SHARMA CU240250980
```

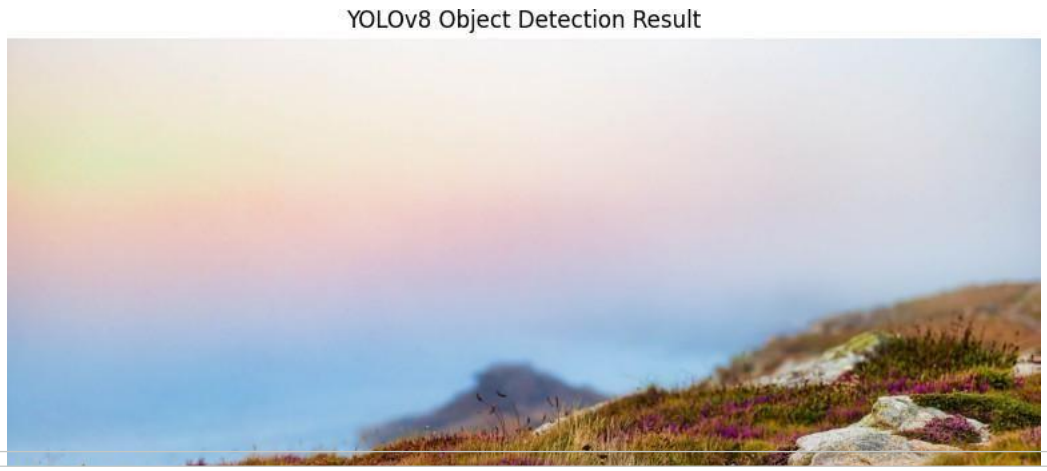
```
# 8. TOTAL OBJECTS
print("\n============================================================")
print("TOTAL OBJECTS DETECTED:", object_count)
print("============================================================")
print("PRIYANSHUN SHARMA CU240250980")
```

```
============================================================
TOTAL OBJECTS DETECTED: 0
============================================================
PRIYANSHUN SHARMA CU240250980
```

```
# 9. DISPLAY ORIGINAL IMAGE
plt.figure(figsize=(10, 7))
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")
plt.show()
```




```
# 10. DISPLAY DETECTION RESULT
plt.figure(figsize=(10, 7))
plt.imshow(detected_image_rgb)
plt.title("YOLOv8 Object Detection Result")
plt.axis("off")
plt.show()
```




```
# 11. SAVE DETECTION RESULT
output_path = r"D:\machine learning\computer vision\detected_result.jpg"
cv2.imwrite(
    output_path,
    detected_image
)
print("\nDetection result saved successfully.")
print("Saved at:", output_path)
print("PRIYANSHUN SHARMA CU240250980")
```

```
Detection result saved successfully.
Saved at: D:\machine learning\computer vision\detected_result.jpg
PRIYANSHUN SHARMA CU240250980
```

```
# 12. EVALUATION
# ============================================================
print("\n============================================================")
print("EVALUATION")
print("============================================================")
print("Number of detected objects:", object_count)
print("\nThe detected objects are displayed with:")
print("1. Bounding boxes")
print("2. Class labels")
print("3. Confidence scores")
print("PRIYANSHUN SHARMA CU240250980")
print("\nExperiment completed successfully.")
```

```

============================================================
EVALUATION
============================================================
Number of detected objects: 0

The detected objects are displayed with:
1. Bounding boxes
2. Class labels
3. Confidence scores
PRIYANSHUN SHARMA CU240250980

Experiment completed successfully.
```